is represented by a line running from the vertex representing that $\mathscr{H}(x)$ upwards out of the diagram, carrying an arrow pointed upwards.
(b) The pairing of a final antiparticle with a field in one of the $\mathscr{H}(x)$ is also represented by a line running from the vertex representing that $\mathscr{H}(x)$ upwards out of the diagram, but carrying an arrow pointed downwards. (Arrows are omitted throughout for particles like γ, π^0, etc. that are their own antiparticles.)
(c) The pairing of an initial particle with a field in one of the $\mathscr{H}(x)$ is represented by a line running into the diagram from below, ending in the vertex representing that $\mathscr{H}(x)$, carrying an arrow pointed upwards.
(d) The pairing of an initial antiparticle with a field in one of the $\mathscr{H}(x)$ is also represented by a line running into the diagram from below, ending in the vertex representing that $\mathscr{H}(x)$, but carrying an arrow pointed downwards.
(e) The pairing of a final particle or antiparticle with an initial particle or antiparticle is represented by a line running clear through the diagram from bottom to top, not touching any vertex, with an arrow pointed upwards or downwards for particles or antiparticles, respectively.
(f) The pairing of a field in $\mathscr{H}(x)$ with a field adjoint in $\mathscr{H}(y)$ is represented by a line joining the vertices representing $\mathscr{H}(x)$ and $\mathscr{H}(y)$, carrying an arrow pointing from y to x.

Note that arrows always point in the direction a particle is moving, and opposite to the direction an antiparticle is moving. (As mentioned above, arrows should be omitted for particles like photons that are their own antiparticles.) The arrow direction indicated in rule (f) is consistent with this convention because a field adjoint in $\mathscr{H}_j(y)$ can either create a particle destroyed by a field in $\mathscr{H}_i(x)$, or destroy an antiparticle created by a field in $\mathscr{H}_i(x)$. Note also that since every field or field adjoint in $\mathscr{H}_i(x)$ must be paired with something, the total number of lines at a vertex of type i, corresponding to a term $\mathscr{H}_i(x)$ in Eq. (6.1.2), is just equal to the total number of field or field adjoint factors in $\mathscr{H}_i(x)$. Of these lines, the number with arrows pointed into the vertex or out of it equals the number of fields or field adjoints respectively in the corresponding interaction term.

To calculate the contribution to the S-matrix for a given process, of a given order N_i in each of the interaction terms $\mathscr{H}_i(x)$ in Eq. (6.1.2), we must carry out the following steps:

(i) Draw all Feynman diagrams containing N_i vertices of each type i, and containing a line coming into the diagrams from below for each particle or antiparticle in the initial state, and a line going upwards out of the diagram for every particle or antiparticle in the final state, together